

Liquid-Sequencer: A Lightweight Liquid Neural Network for Real-Time Fetal Congenital Heart Disease Diagnosis

Sibo Qiao, Ju Gao, Min Wang, Qiang Guo, Cong Liu, Shuqiang Wang, Zhiyuan Zhao, Mohammad Shabaz

Abstract—Congenital heart disease (CHD) is the leading cause of neonatal mortality worldwide, making early and accurate diagnosis crucial. In resource-constrained regions, standardized ultrasound screening remains difficult due to the shortage of specialized clinicians. Digital Twin (DT) technology, which constructs virtual AI diagnostic models, enables personalized assessments of heart structure and function, offering intelligent diagnostic support in primary or remote healthcare settings. However, the real-time updates required by DT systems place increased demands on the inference speed and computational efficiency of AI models. Existing methods often suffer from parameter redundancy and inference delays, making them inadequate for meeting the low-latency, large-scale needs of clinical applications. To overcome these challenges, we propose the Liquid-Sequencer, a lightweight model for the diagnosis of fetal CHD. The model first employs a convolutional network with DPSE (Depthwise Separable and Squeeze-and-Excitation) modules to efficiently extract spatial features by utilizing depthwise separable convolution and channel attention. These feature maps are then processed by a bidirectional liquid sequence module, where orthogonally scanned Liquid Neural Networks (LNNs) capture global context with linear complexity, offering a more efficient alternative to self-attention mechanisms. This integration of spatial and sequential learning is well-suited to the dynamic nature of fetal cardiac ultrasound and the demands of DT systems. Experimental results on 12 datasets demonstrate

outstanding performance with only 0.30M parameters. Additionally, t-SNE visualizations reveal highly discriminative feature representations and clear inter-class separations, underscoring the model's potential as an advanced diagnostic tool.

Index Terms—Fetal CHD diagnosis, Lightweight model, Liquid neural networks, Digital Twin.

I. INTRODUCTION

Congenital heart disease (CHD) stands as one of the most common and severe types of birth defects, affecting approximately 6 to 12 per 1,000 live births worldwide, and represents the leading cause of neonatal mortality [1]. Beyond its clinical severity, CHD imposes a substantial psychological and economic burden on both families and healthcare systems. Therefore, timely and accurate prenatal diagnosis is essential for enabling early intervention, improving prognosis, and reducing neonatal mortality.

Despite its obvious importance, early CHD screening still faces numerous challenges. For example, fetal echocardiography, as the primary non-invasive method for prenatal CHD diagnosis, heavily relies on the operator's professional skills to accurately capture the complex structure and dynamic changes of the fetal heart. However, in practice, the accuracy of ultrasound diagnostics is often limited by factors such as variations in operator experience, accessibility to specialized physicians, and constraints of high-performance computing platforms. These factors lead to delays in screening, inconsistent results, and other issues, which have become commonplace and hinder the widespread adoption and standardization of early CHD diagnosis, highlighting the urgent need for efficient, accurate, and portable intelligent diagnostic solutions.

The advancement of deep learning has spurred significant progress in medical image analysis across various modalities, including X-ray, computed tomography (CT), and magnetic resonance imaging (MRI). Architectures such as Convolutional Neural Networks (CNNs) [2] and attention-based models like the Transformer [3]–[5] have markedly advanced disease diagnosis by enhancing spatial feature representation and modeling global dependencies. For instance, Perera et al. [4] introduced SegFormer3D, which achieves competitive performance with a substantially reduced parameter count. Similarly, SMAFormer, proposed by Zheng et al. [5], effectively captures both local and global features for multi-organ segmentation. More recent

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studies have demonstrated that hybrid CNN-Transformer models [6]–[8], self-supervised pre-training [9], [10], and semi-supervised learning [11]–[13] can enhance model robustness and generalization, particularly when annotated data is scarce. However, these methods are often computationally intensive, which hinders their practical application and deployment in resource-constrained clinical environments.

Deep learning has become a cornerstone technology for the intelligent analysis of fetal echocardiography, yielding significant progress in areas such as the automatic selection of standard views [14]–[17], segmentation of anatomical structures including the heart and great vessels [5], [18]–[20], and the early screening and diagnosis of CHD [21]–[26], all of which demonstrate promising clinical potential. For example, Krishna et al. [15] used an ensemble of CNNs for robust detection of six standard fetal views. Yu et al. [18] introduced MLCL-CM, which improves four-chamber segmentation by exploiting unlabeled intermediate frames and contrastive learning.

In the domain of CHD diagnosis, Qiao et al. first introduced RLDS [22], a residual learning-based model that extracts the cardiac region of interest from the four-chamber view using candidate regions from their prior FLDS model [21] to perform preliminary CHD screening. Building on this, they proposed SPReCHD [23], a dual-branch network for semantic parsing of the four-chamber view, which achieves accurate CHD diagnosis through multi-scale modeling of the fetal cardiac structure. Gao et al. [25] designed a multi-scale feature fusion method using a Gated Swin-Transformer, alleviating class imbalance with data augmentation and emphasizing salient features through dynamic weighting. Lu et al. [24] introduced SKGC, a three-stage diagnostic model that integrates semantic knowledge extraction, multi-feature fusion, and a classifier-enhanced decision module. Saha et al. [27] proposed STUD, a privacy-preserving zero-shot diagnosis framework that treats CHD as an anomaly detection task using normal samples and enables multi-center model fusion without data sharing. Guan et al. [26] further developed DPC-Net, a dual-path complementary network for efficient feature extraction and segment-level phonocardiogram image classification without cardiac cycle segmentation, with occlusion maps validating its design.

Despite significant progress in intelligent fetal ultrasound analysis, existing studies still have difficulty modeling the dynamic relationship between cardiac structure and function, which limits stable, interpretable, and generalizable diagnostic decisions. Digital Twin (DT) technology offers a promising direction for precise early CHD diagnosis. By integrating multidimensional physiological information, DT can build a dynamic correspondence between structure and function, forming a personalized digital replica of the fetal heart through AI diagnostic models and enabling dynamic characterization of structural-functional changes. However, DT systems place strict demands on real-time performance and lightweight model design. Current methods often contain redundant parameters and exhibit inference latency, which restricts efficient deployment and clinical use in resource-constrained settings.

Researchers have proposed various lightweight neural net-

work architectures, such as MobileNet [28], [29], ShuffleNet [30], [31], and EfficientNet [32], which reduce model size while significantly accelerating inference speed. Recent studies have also explored strategies such as knowledge distillation [33], [34], search for neural architecture [35], [36], and task-specific fine-tuning [37], [38] to improve computational efficiency without substantial performance trade-off. However, these lightweight approaches remain limited in fetal echocardiography because they often fail to capture subtle structural abnormalities and dynamic cardiac motion that are critical for reliable CHD diagnosis. In addition, clinical datasets are small and imbalanced, which increases the risk of overfitting and reduces sensitivity to rare CHD subtypes. Practical deployment, therefore, requires models with high diagnostic accuracy, strong temporal representation, and low computational cost, yet current methods still struggle to meet all these requirements.

Liquid Neural Networks (LNNs), a biologically inspired paradigm, are defined by models built on continuous-time dynamics [39]. Unlike traditional static architectures, LNNs adaptively adjust internal states to dynamic input changes, selectively enhancing relevant signals while suppressing noise. This affords strong temporal modeling and robustness to interference, and studies show that in noisy navigation tasks, LNNs outperform conventional recurrent networks [40]. Although their use in medical imaging remains early [41]–[43], their strengths in capturing temporal consistency and efficient noise suppression highlight strong potential for ultrasound video analysis.

To address the aforementioned challenges, we introduce the Liquid-Sequencer, a lightweight framework for fetal CHD diagnosis that combines the spatial feature extraction capabilities of CNNs with the temporal dynamic modeling of LNNs. The framework utilizes a compact convolutional module to efficiently capture key spatial features, while a liquid block adaptively models the dynamic changes within ultrasound sequences. By integrating axial processing with bidirectional feature optimization, the model effectively encodes structural features such as ventricular septal boundaries and outflow tracts, while remaining robust to artifacts. Our proposed Liquid-Sequencer achieves high-precision CHD diagnosis while significantly reducing parameter count and computational demands, enabling personalized virtual heart modeling and real-time diagnostic optimization. Moreover, the lightweight design of the model facilitates its deployment on portable devices, paving the way for the clinical adoption of digital twin technology in resource-limited settings. Our primary contributions are as follows:

(1) We propose Liquid-Sequencer, an efficient hybrid visual architecture, achieving an effective synergy between spatial and sequential information processing. This is accomplished by first extracting high-quality local spatial features using deep convolution and then feeding them into LNNs for global sequence modeling.

(2) We design a novel bidirectional liquid sequence processing module. This module decomposes the learning of 2D spatial dependencies into parallel horizontal and vertical one-dimensional scans, efficiently capturing anisotropic global

context with linear computational complexity and offering a lightweight yet powerful alternative to self-attention mechanisms.

(3) We validate the model's state-of-the-art (SOTA) performance on a real-world fetal cardiac ultrasound dataset, showing that the Liquid-Sequencer significantly outperforms mainstream lightweight models in diagnostic accuracy while maintaining low computational overhead and memory footprint. This balance of accuracy and efficiency makes it well-suited for deployment on resource-constrained medical devices and integration into DT systems.

II. METHODS

We propose the Liquid-Sequencer, a lightweight and efficient deep learning architecture for fetal CHD diagnosis from ultrasound images. The model's design is predicated on a synergistic fusion of convolutional feature extraction and a novel sequential processing mechanism based on LNNs. The architecture is constructed around a feature extractor composed of a stack of **DeP**thwise separable and **S**queeze-and-**E**xcitation (DPSE) blocks, which yields rich spatial representations. These are subsequently processed by a core Liquid Block that leverages the continuous-time dynamics of LNNs to model complex spatial dependencies. This section first elucidates the theoretical underpinnings of Liquid Time-Constant (LTC) Networks, and then presents a detailed exposition of the Liquid-Sequencer's architectural design.

A. Liquid Time-Constant Networks

LTC networks represent a paradigm shift from conventional discrete-time Recurrent Neural Networks (RNNs), drawing inspiration from the neural dynamics of simple organisms like *C. elegans*. LTCs model the hidden state dynamics of a neural population using a system of coupled Ordinary Differential Equations (ODEs), enabling them to process continuous-time signals and adapt their behavior on the fly. The evolution of the hidden state vector $\mathbf{x}(t) \in \mathbb{R}^N$ is governed by:

$$\frac{d\mathbf{x}(t)}{dt} = -\frac{\mathbf{x}(t)}{\tau} + \mathbf{S}(t), \quad (1)$$

where $\tau \in \mathbb{R}^N$ is a vector of intrinsic neural time constants, and $\mathbf{S}(t) \in \mathbb{R}^N$ represents the synaptic input current. This input is defined as a state- and input-dependent gated interaction:

$$\mathbf{S}(t) = f(\mathbf{x}(t), \mathbf{I}(t), t, \theta), (A - \mathbf{x}(t)), \quad (2)$$

where $f(\cdot)$ is a compact neural network parameterized by learnable weights θ , $\mathbf{I}(t)$ is the external input feature at time t , and A is a constant representing the synaptic reversal potential. Substituting $\mathbf{S}(t)$ back, we obtain the full state dynamics equation for LTCs:

$$\frac{d\mathbf{x}(t)}{dt} = -\left(\frac{1}{\tau} + f(\mathbf{x}(t), \mathbf{I}(t), t, \theta)\right)\mathbf{x}(t) + f(\mathbf{x}(t), \mathbf{I}(t), t, \theta)A. \quad (3)$$

A key property of this formulation is that the neural mapping $f(\cdot)$ not only modulates the state derivative but also

dynamically controls the effective system time constant, τ_{sys} , for each neuron:

$$\tau_{sys} = \frac{\tau}{1 + \tau f(\mathbf{x}(t), \mathbf{I}(t), t, \theta)}. \quad (4)$$

This mechanism allows LTC networks to adaptively adjust their temporal integration scales. They can dwell on salient features by increasing their time constant and accelerating through less informative segments, a critical capability for processing the variable-rate dynamics inherent in ultrasound video sequences.

To enable practical implementation on digital hardware, the continuous-time dynamics of Eq. 3 must be discretized. We employ a highly efficient and stable closed-form solver, which can be interpreted as an Implicit-Explicit (IMEX) scheme. This solver yields the following update rule for a discrete time step Δt :

$$\mathbf{x}(t + \Delta t) = \frac{\mathbf{x}(t) + \Delta t, f(\mathbf{x}(t), \mathbf{I}(t), t, \theta)A}{1 + \Delta t \left(\frac{1}{\tau} + f(\mathbf{x}(t), \mathbf{I}(t), t, \theta)\right)}. \quad (5)$$

Crucially, unlike traditional iterative ODE solvers (e.g., Runge-Kutta or variable-step methods) that involve multiple function evaluations per step, this closed-form solution computes the state transition in a single, direct pass involving only basic arithmetic operations. The denominator of this equation acts as an adaptive normalization factor, dynamically regulating the magnitude of the state update and ensuring numerical stability. Consequently, the inference latency is determined solely by the network depth rather than the solver's complexity. This grants LTCs the expressive power of continuous-time models while maintaining the computational efficiency and constant time complexity of conventional RNNs, making them exceptionally suitable for real-time diagnostic applications.

B. The Liquid-Sequencer Architecture

The overall architecture of the Liquid-Sequencer, depicted in Fig. 1, is a hierarchical network. It first employs our proposed DPSE Block to extract spatial features from the input ultrasound image. The core of the model is the Liquid Block, which transforms these spatial features into sequences and processes them using bidirectional LNNs to capture long-range contextual dependencies. The following subsections detail the design of each constituent module.

1) DPSE Block: The DPSE Block, an acronym for DePthwise separable and Squeeze-and-Excitation Block, is a composite module engineered for an optimal trade-off between computational efficiency and representational power. Given an input tensor $Z \in \mathbb{R}^{H \times W \times C}$, the block first performs depthwise separable convolution. The depthwise convolution operation, which applies a single filter per input channel, can be expressed for the c -th channel at spatial location (i, j) as:

$$Z'_{dw}[i, j, c] = \sum_{k,l} K_{dw}[k, l, c] \cdot Z[i+k, j+l, c], \quad (6)$$

where $K_{dw} \in \mathbb{R}^{K_h \times K_w \times C}$ is the depthwise kernel. This is followed by a pointwise convolution, a 1×1 convolution that

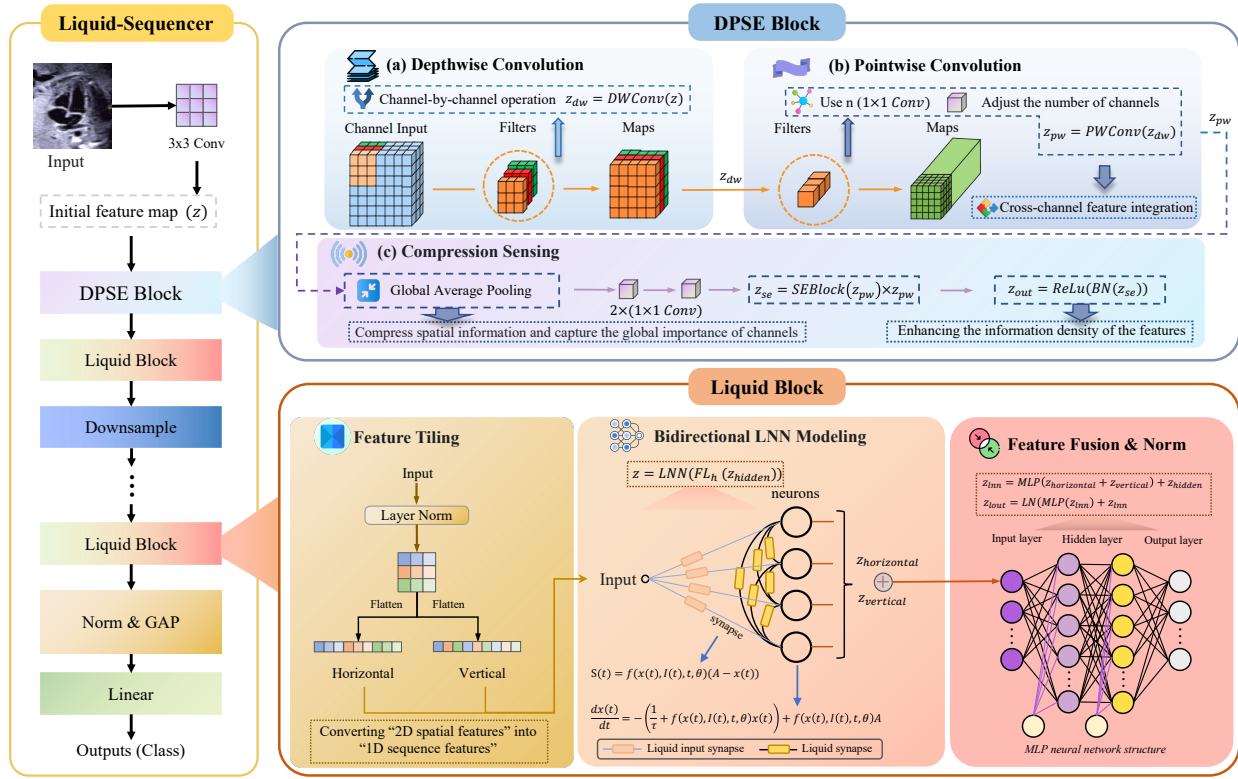


Fig. 1. Framework Diagram of the Liquid-Sequencer Model: First, the model employs a 3×3 convolution to map the input image into feature z . Second, feature z is compressed into the depth feature z_{out} via depthwise separable convolution. Subsequently, the depth feature is fed into multiple Liquid Blocks and undergoes computation through the bidirectional liquid neural network; finally, the classification result is obtained through global average pooling and linear mapping.

TABLE I
DATASETS USED IN THIS STUDY.

Name	Data Modality	Task (Classes)	Training/Test	Image Size	Availability
CHD-US	Fetal Cardiac Ultrasound	Binary (2)	4566 / 1000	$128 \times 128 \times 3$	Private
Fetal-Planes-DB [44]	Maternal-Fetal Ultrasound	Multi-Class (6)	7129 / 5271	$128 \times 128 \times 3$	Public
CPN X-ray [45], [46]	Chest X-Ray	Multi-Class (3)	3838 / 1390	$128 \times 128 \times 3$	Public
BloodMNIST [47]	Blood Cell Microscopy	Multi-Class (8)	11959 / 3421	$28 \times 28 \times 3$	Public
BreastMNIST [47]	Breast Ultrasound	Binary (2)	546 / 156	$28 \times 28 \times 1$	Public
PathMNIST [47]	Colon Pathology	Multi-Class (9)	89996 / 7180	$28 \times 28 \times 3$	Public
PneumoniaMNIST [47]	Chest X-Ray	Binary (2)	4708 / 624	$28 \times 28 \times 1$	Public

linearly combines the channels:

$$Z_{pw}[i, j, c'] = \sum_{c=1}^C K_{pw}[1, 1, c, c'] \cdot Z'_{dw}[i, j, c]. \quad (7)$$

The resulting feature map $Z_{pw} \in \mathbb{R}^{H \times W \times C'}$ undergoes channel-wise recalibration via the Squeeze-and-Excitation mechanism. The Squeeze operation aggregates global spatial information into a channel descriptor $s \in \mathbb{R}^{C'}$:

$$s_{c'} = \mathcal{F}_{sq}(Z_{pw})c' = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W Z_{pw}[i, j, c']. \quad (8)$$

The Excitation operation then generates channel-wise attention weights $\alpha \in \mathbb{R}^{C'}$ by processing s through two fully-connected layers:

$$\alpha = \mathcal{F}_{ex}(s) = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 s)), \quad (9)$$

where $\mathbf{W}_1 \in \mathbb{R}^{\frac{C'}{r} \times C'}$ and $\mathbf{W}_2 \in \mathbb{R}^{C' \times \frac{C'}{r}}$ are weight matrices, δ is the ReLU activation, and σ is the sigmoid function. The final output of the DPSE block, Z_{out} , is obtained by applying these attention weights, followed by Batch Normalization ($\mathcal{BN}$) and ReLU activation:

$$Z_{out} = \delta(\mathcal{BN}(\alpha \odot Z_{pw})), \quad (10)$$

where $\odot$ denotes channel-wise multiplication.

In our architecture, the feature extractor begins with an initial strided convolution. This is followed by a stack of three DPSE blocks, with a strided convolution between each block. This results in a final feature map, $Z_{hidden} \in \mathbb{R}^{H/16 \times W/16 \times C'}$, where $C' = 64$.

2) Bidirectional Liquid Block: This block is designed to efficiently capture spatial dependencies across the feature map $Z_{in} \in \mathbb{R}^{B \times H' \times W' \times C'}$. For the first block, Z_{in} is the output Z_{hidden} from the DPSE module. Instead of full serialization, we adopt a highly parallelized approach by treating the rows

and columns of the feature map as independent sequences. The process is divided into two parallel paths: a horizontal path and a vertical path, each equipped with its own dedicated LNN module to learn direction-specific patterns.

1. Horizontal Processing Path: To capture dependencies along the horizontal axis, we treat each of the H' rows as an independent sequence of length W' . The input tensor is first reshaped from $\mathbb{R}^{B \times H' \times W' \times C'}$ to $\mathbb{R}^{(B \cdot H') \times W' \times C'}$, grouping all rows from all samples in the batch together. This creates a new, larger batch of size $(B \cdot H')$.

$$Z_{hseq} = \text{Reshape}(Z_{in}). \quad (11)$$

A dedicated bidirectional LNN for horizontal processing, denoted LNN_{θ_h} , is then applied to model intra-row relationships. This operation processes the tensor, yielding an output of shape $\mathbb{R}^{(B \cdot H') \times W' \times C''}$, where the channel dimension may change from C' to C'' .

$$Z'_{hseq} = \text{LNN}_{\theta_h}(Z_{hseq}). \quad (12)$$

Finally, the processed tensor is reshaped from $\mathbb{R}^{(B \cdot H') \times W' \times C''}$ back to its original 2D spatial arrangement of $\mathbb{R}^{B \times H' \times W' \times C''}$ to yield the horizontally-aware feature map $Z_{horizontal}$.

$$Z_{horizontal} = \text{Reshape}(Z'_{hseq}). \quad (13)$$

2. Vertical Processing Path: To capture vertical dependencies, we perform a similar operation along the columns. First, the spatial dimensions of the input tensor Z_{in} are transposed from $\mathbb{R}^{B \times H' \times W' \times C'}$ to $\mathbb{R}^{B \times W' \times H' \times C'}$, bringing the vertical dimension (H') into the sequence position.

$$Z_{vpermuted} = \text{Permute}(Z_{in}). \quad (14)$$

Now, treating each of the W' columns as a sequence of length H' , we reshape the tensor from $\mathbb{R}^{B \times W' \times H' \times C'}$ to $\mathbb{R}^{(B \cdot W') \times H' \times C'}$.

$$Z_{vseq} = \text{Reshape}(Z_{vpermuted}). \quad (15)$$

A separate bidirectional LNN for vertical processing, LNN_{θ_v} , processes these column-sequences. The output is then reshaped and permuted back to the original $H' \times W'$ layout to produce $Z_{vertical}$.

3. Fusion and Refinement: The feature maps from both paths, now containing specialized horizontal and vertical context, are fused via element-wise addition and processed by an MLP, followed by a residual connection. This is then refined by an FFN block:

$$\begin{aligned} Z_{l_{nn}} &= \Phi_{MLP}(Z_{horizontal} + Z_{vertical}) + Z_{in}, \\ Z_{l_{out}} &= \text{LN}(\Psi_{FFN}(Z_{l_{nn}})) + Z_{l_{nn}}, \end{aligned} \quad (16)$$

where Φ_{MLP} is implemented using 1×1 convolutions, and $\Psi_{FFN}(X) = \mathbf{W}_{ffn,2} \cdot \text{GELU}(\mathbf{W}_{ffn,1}X)$ with LN denoting Layer Normalization.

This parallel-path architecture, with dedicated LNNs for orthogonal axes, allows the model to learn distinct and rich spatial features from two perspectives, while the batch-oriented processing ensures computational efficiency. The entire model

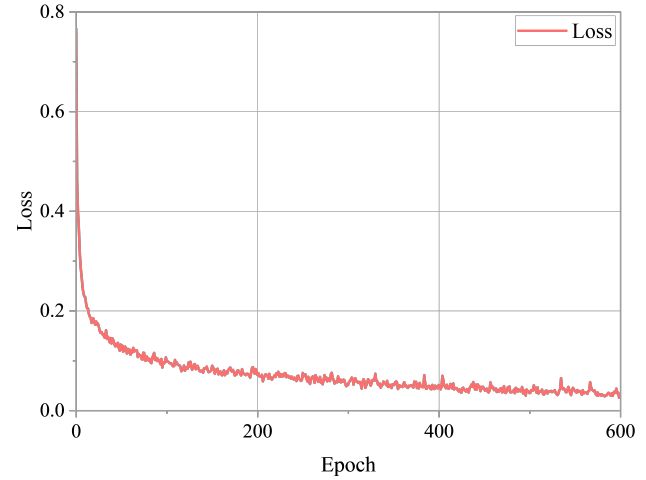


Fig. 2. Model Training Curve

is constructed by stacking these Liquid Sequential Processing Blocks. Critically, this design offers distinct advantages over conventional spatial analysis benchmarks. Unlike Convolutional Neural Networks which are constrained by local receptive fields and require deep stacking to capture global context, our row-column decomposition enables the LNN to model global long-range dependencies across the entire feature map in a single layer. Furthermore, compared to discrete recurrent units, the intrinsic continuous-time dynamics of the Liquid Neural Network provide superior numerical stability and expressivity, particularly when modeling complex non-linear spatial transitions. By decoupling the spatial dimensions, our approach also circumvents the quadratic computational complexity of Global Self-Attention mechanisms and the sequential bottlenecks of pixel-wise 2D-RNNs, thereby achieving a synergy of linear linear scalability and robust representation learning.

3) Classification Head: After passing through the stack of LNN blocks, the final feature map is processed by a classification head. We first apply global average pooling to the spatial dimensions of the feature map to produce a single feature vector per image. This vector is then passed through a Layer Normalization layer, followed by a final fully-connected linear layer that outputs the logits for the C classes.

$$Y_{logits} = \text{FC}(\text{LN}(\text{GAP}(Z_{final}))) \quad (17)$$

The entire model is trained end-to-end using a standard cross-entropy loss function.

III. EXPERIMENTS AND RESULTS

A. Datasets

We evaluate our model using a proprietary fetal cardiac ultrasound dataset (CHD-US) and 11 public datasets to assess generalization across medical and natural imaging tasks. Table I details the modalities, categories, and splits for all datasets.

CHD-US. CHD-US is a proprietary fetal cardiac ultrasound dataset that we constructed, consisting of 5,566 fetal four-chamber ultrasound images. Among them, 2,745 images are of CHD cases, while 2,821 images are of healthy fetal hearts.

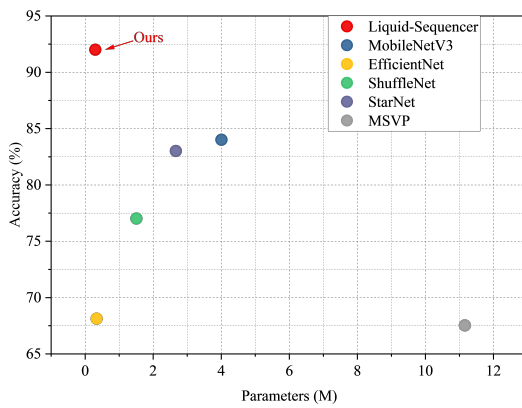


Fig. 3. Performance-Parameter Trade-off Plot

TABLE II
PARAMETERS

Model	Parameters(M)	MACs(M)
Liquid-Sequencer(our)	0.30	40.22
MobileNetV3 [29]	4.01	135.57
EfficientNet [32]	0.34	14.22
ShuffleNet [30]	1.51	20.77
StarNet [48]	2.67	139.47
MSVP [49]	11.17	557.89

Each image is scanned from a single patient, meaning that each image corresponds to one patient. We use 4,566 images as the training set and 1,000 images as the test set, with no overlap between the training and testing sets. These images are collected from Qingdao Women and Children's Hospital and annotated by three professional ultrasound physicians. The use of this dataset was approved by the Institutional Review Board (ID: QFELL-KY-2024-64).

Fetal-Planes-DB [44]. This public dataset contains 12,400 maternal-fetal screening images across six anatomical classes (e.g., abdomen, brain, thorax). We utilize 7,129 images for training and 5,271 for testing.

CPN X-ray [45], [46]. This dataset includes 5,228 chest X-rays categorized into COVID-19, pneumonia, and normal classes. The split consists of 3,838 training and 1,390 test samples.

MedMNIST [47]. We select six standardized 2D biomedical classification datasets from the MedMNIST collection: BloodMNIST, BreastMNIST, PathMNIST, and PneumoniaMNIST. These cover diverse modalities including X-ray, OCT, and pathology slides.

B. Implementation Details

We select several commonly used lightweight vision classification networks to validate the effectiveness and competitiveness of our proposed model in lightweight design and fetal congenital heart disease ultrasound image classification tasks. To ensure fair comparisons, we apply unified preprocessing to the datasets before training each network. Since the proposed model is insensitive to image size, we scale the images to different sizes and normalize their pixel values to the range $[-1, 1]$. Additionally, to enhance the model's robustness, we

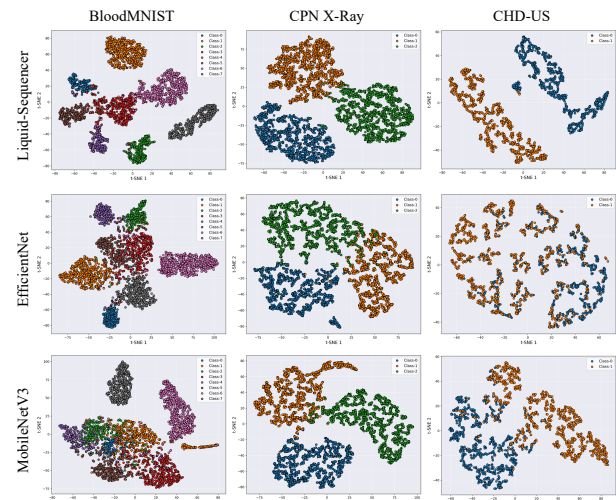


Fig. 4. t-SNE Visualization Results of Various Models on BloodMNIST, CPN X-Ray, and CHD-US Datasets

apply random horizontal and vertical flips to the images with a probability of $p=0.5$, as well as random rotations of 15° with a probability of $p=0.3$. The image sizes for each dataset are listed in Table I.

During the training process, we use the AdamW optimizer to optimize the model parameters, with an initial learning rate of 5×10^{-3} . The learning rate is adjusted using the ExponentialLR scheduler with a decay factor (γ) of 0.999. Each model is trained for 200 epochs with a batch size of 128. To improve training stability, we employ an Exponential Moving Average (EMA) approach to update the model parameters. All training is conducted on a server equipped with an Intel(R) Xeon(R) Gold 6330 processor and an NVIDIA A100-SXM4-40GB GPU. We adopt accuracy, precision, recall, F1-Score, and ROC-AUC as evaluation metrics.

C. Results and Discussion

1) Model Training Curve: Figure 2 illustrates the training loss progression over 600 epochs. Aggregating loss per epoch yields a smooth trajectory that clarifies convergence behavior. The model demonstrates robust learning: an initial sharp decline reduces the loss from 0.76 to 0.20 within the first 20 epochs, indicating rapid feature assimilation. Subsequently, the curve follows a steady, monotonic decrease with minimal variance, stabilizing at approximately 0.03 by epoch 600. This confirms effective parameter optimization and convergence to a stable solution.

2) Model Parameter Comparison: We evaluate the parameter scale and computational cost of the proposed Liquid-Sequencer against representative lightweight models: MobileNetV3 [29], EfficientNet [32], ShuffleNet [30], StarNet [48], and MSVP [49]. Table II summarizes the model parameters and multiply-accumulate operations (MACs).

The results indicate that Liquid-Sequencer achieves a highly efficient parameter size of 0.30M. This represents a 92.5% reduction compared to MobileNetV3 (4.01M), 80.1% compared to ShuffleNet (1.51M), and a massive reduction compared to the heavy MSVP model (11.17M). Even against the

TABLE III
COMPARISON OF CLASSIFICATION PERFORMANCE ACROSS DIFFERENT DATASETS FOR VARIOUS MODELS

Dataset	Model	Accuracy(%) $\uparrow$	Precision(%) $\uparrow$	Recall(%) $\uparrow$	F1(%) $\uparrow$	AUC $\uparrow$
CHD-US	Liquid-Sequencer	92.0 $\pm$ 0.4	90.1 $\pm$ 0.3	90.0 $\pm$ 0.3	89.9 $\pm$ 0.3	0.968 $\pm$ 0.005
	MobileNetV3 [29]	84.0 $\pm$ 0.2	84.2 $\pm$ 0.1	84.0 $\pm$ 0.01	83.9 $\pm$ 0.1	0.901 $\pm$ 0.012
	EfficientNet [32]	68.1 $\pm$ 0.2	69.4 $\pm$ 0.3	68.8 $\pm$ 0.01	67.4 $\pm$ 0.01	0.741 $\pm$ 0.007
	ShuffleNet [30]	77.0 $\pm$ 0.3	76.9 $\pm$ 0.3	76.9 $\pm$ 0.3	76.9 $\pm$ 0.2	0.868 $\pm$ 0.056
	StarNet [48]	83.0 $\pm$ 0.1	83.9 $\pm$ 0.1	84.3 $\pm$ 0.1	83.0 $\pm$ 0.1	0.896 $\pm$ 0.008
	MSVP [49]	67.5 $\pm$ 0.2	64.8 $\pm$ 0.2	75.6 $\pm$ 0.1	69.8 $\pm$ 0.1	0.749 $\pm$ 0.003
FetalDB [44]	Liquid-Sequencer	90.6 $\pm$ 0.1	89.7 $\pm$ 0.1	89.2 $\pm$ 0.2	89.4 $\pm$ 0.1	0.998 $\pm$ 0.003
	MobileNetV3 [29]	83.7 $\pm$ 0.1	82.9 $\pm$ 0.2	80.1 $\pm$ 0.1	80.1 $\pm$ 0.1	0.963 $\pm$ 0.004
	EfficientNet [32]	78.0 $\pm$ 0.1	75.4 $\pm$ 0.2	69.0 $\pm$ 0.2	70.8 $\pm$ 0.2	0.929 $\pm$ 0.011
	ShuffleNet [30]	83.6 $\pm$ 0.1	84.4 $\pm$ 0.1	77.6 $\pm$ 0.1	80.0 $\pm$ 0.1	0.968 $\pm$ 0.003
	StarNet [48]	83.9 $\pm$ 0.2	82.1 $\pm$ 0.2	79.1 $\pm$ 0.1	80.2 $\pm$ 0.2	0.973 $\pm$ 0.005
	MSVP [49]	86.6 $\pm$ 0.1	80.9 $\pm$ 0.2	87.2 $\pm$ 0.1	83.0 $\pm$ 0.2	0.978 $\pm$ 0.001
CPN [45] [46]	Liquid-Sequencer	99.2 $\pm$ 0.2	99.2 $\pm$ 0.3	99.3 $\pm$ 0.1	99.2 $\pm$ 0.2	0.999 $\pm$ 0.001
	MobileNetV3 [29]	98.7 $\pm$ 0.2	98.8 $\pm$ 0.2	98.7 $\pm$ 0.2	98.6 $\pm$ 0.2	0.999 $\pm$ 0.001
	EfficientNet [32]	97.1 $\pm$ 0.2	97.1 $\pm$ 0.1	97.3 $\pm$ 0.1	97.1 $\pm$ 0.1	0.992 $\pm$ 0.002
	ShuffleNet [30]	98.2 $\pm$ 0.1	98.2 $\pm$ 0.2	98.2 $\pm$ 0.1	98.1 $\pm$ 0.2	0.997 $\pm$ 0.001
	StarNet [48]	99.2 $\pm$ 0.1	99.1 $\pm$ 0.1	99.2 $\pm$ 0.1	99.2 $\pm$ 0.2	0.999 $\pm$ 0.001
	MSVP [49]	93.8 $\pm$ 0.1	93.8 $\pm$ 0.2	93.8 $\pm$ 0.3	93.8 $\pm$ 0.1	0.991 $\pm$ 0.003
BloodMNIST [47]	Liquid-Sequencer	96.4 $\pm$ 0.05	96.1 $\pm$ 0.06	96.1 $\pm$ 0.09	96.1 $\pm$ 0.07	0.998 $\pm$ 0.001
	MobileNetV3 [29]	95.2 $\pm$ 0.07	94.6 $\pm$ 0.1	94.8 $\pm$ 0.07	94.6 $\pm$ 0.09	0.997 $\pm$ 0.001
	EfficientNet [32]	95.4 $\pm$ 0.04	94.9 $\pm$ 0.06	95.0 $\pm$ 0.06	94.9 $\pm$ 0.06	0.997 $\pm$ 0.001
	ShuffleNet [30]	94.6 $\pm$ 0.05	93.8 $\pm$ 0.06	93.9 $\pm$ 0.08	93.8 $\pm$ 0.06	0.996 $\pm$ 0.001
	StarNet [48]	95.6 $\pm$ 0.05	95.6 $\pm$ 0.07	95.1 $\pm$ 0.04	95.3 $\pm$ 0.05	0.997 $\pm$ 0.001
	MSVP [49]	96.2 $\pm$ 0.1	95.9 $\pm$ 0.1	96.0 $\pm$ 0.2	95.9 $\pm$ 0.1	0.998 $\pm$ 0.002
BreastMNIST [47]	Liquid-Sequencer	86.5 $\pm$ 0.06	82.9 $\pm$ 0.1	83.4 $\pm$ 0.05	83.0 $\pm$ 0.06	0.876 $\pm$ 0.040
	MobileNetV3 [29]	80.1 $\pm$ 0.1	75.5 $\pm$ 0.2	76.8 $\pm$ 0.03	75.7 $\pm$ 0.1	0.857 $\pm$ 0.001
	EfficientNet [32]	76.3 $\pm$ 0.06	70.4 $\pm$ 0.3	61.9 $\pm$ 0.1	63.2 $\pm$ 0.1	0.732 $\pm$ 0.038
	ShuffleNet [30]	85.2 $\pm$ 0.06	81.6 $\pm$ 0.1	81.8 $\pm$ 0.1	81.4 $\pm$ 0.04	0.875 $\pm$ 0.012
	StarNet [48]	84.6 $\pm$ 0.2	81.3 $\pm$ 0.4	78.1 $\pm$ 0.3	79.4 $\pm$ 0.3	0.877 $\pm$ 0.002
	MSVP [49]	63.6 $\pm$ 0.1	75.0 $\pm$ 0.1	78.5 $\pm$ 0.1	76.7 $\pm$ 0.2	0.470 $\pm$ 0.006
PathMNIST [47]	Liquid-Sequencer	84.4 $\pm$ 0.1	79.6 $\pm$ 0.1	78.7 $\pm$ 0.1	78.6 $\pm$ 0.2	0.957 $\pm$ 0.007
	MobileNetV3 [29]	78.5 $\pm$ 0.2	74.9 $\pm$ 0.2	75.9 $\pm$ 0.1	74.3 $\pm$ 0.2	0.958 $\pm$ 0.006
	EfficientNet [32]	79.2 $\pm$ 0.1	74.9 $\pm$ 0.1	76.0 $\pm$ 0.2	74.2 $\pm$ 0.1	0.942 $\pm$ 0.006
	ShuffleNet [30]	78.0 $\pm$ 0.1	75.2 $\pm$ 0.1	73.4 $\pm$ 0.1	72.9 $\pm$ 0.1	0.947 $\pm$ 0.007
	StarNet [48]	79.8 $\pm$ 0.1	75.2 $\pm$ 0.1	76.0 $\pm$ 0.1	74.9 $\pm$ 0.1	0.959 $\pm$ 0.006
	MSVP [49]	83.5 $\pm$ 0.1	84.9 $\pm$ 0.1	83.3 $\pm$ 0.1	83.1 $\pm$ 0.2	0.985 $\pm$ 0.002
PneumoniaMNIST [47]	Liquid-Sequencer	88.3 $\pm$ 0.01	89.1 $\pm$ 0.1	86.1 $\pm$ 0.1	87.2 $\pm$ 0.07	0.948 $\pm$ 0.001
	MobileNetV3 [29]	86.7 $\pm$ 0.1	89.8 $\pm$ 0.1	83.1 $\pm$ 0.2	84.9 $\pm$ 0.1	0.943 $\pm$ 0.003
	EfficientNet [32]	82.0 $\pm$ 0.2	83.4 $\pm$ 0.1	78.4 $\pm$ 0.1	79.7 $\pm$ 0.2	0.894 $\pm$ 0.002
	ShuffleNet [30]	86.9 $\pm$ 0.1	87.6 $\pm$ 0.2	84.6 $\pm$ 0.1	85.6 $\pm$ 0.1	0.949 $\pm$ 0.001
	StarNet [48]	87.3 $\pm$ 0.1	91.1 $\pm$ 0.1	83.5 $\pm$ 0.1	85.5 $\pm$ 0.2	0.965 $\pm$ 0.001
	MSVP [49]	76.2 $\pm$ 0.1	76.2 $\pm$ 0.1	81.3 $\pm$ 0.1	78.6 $\pm$ 0.1	0.902 $\pm$ 0.001

efficient EfficientNet (0.34M), our model shows a slight advantage. Regarding computational complexity, Liquid-Sequencer records 40.22M MACs, which is substantially lower than MobileNetV3 (135.57M), StarNet (139.47M), and MSVP (557.89M). Although ShuffleNet and EfficientNet achieve lower MACs, Liquid-Sequencer maintains a competitive complexity range suitable for resource-constrained edge deployment while offering superior accuracy.

Figure 3 visualizes the trade-off between performance (accuracy) and complexity (parameters). Ideally, models appear in the upper-left region, signifying high accuracy with low complexity. Liquid-Sequencer occupies this optimal position, demonstrating a superior balance between efficiency and effectiveness compared to the baselines.

3) *Model Performance Comparison*: We evaluate Liquid-Sequencer across seven medical image datasets. Table III summarizes the results against baseline models.

On the primary medical datasets, Liquid-Sequencer demonstrates superior classification. For CHD-US, it achieves 92.0% accuracy and 0.968 AUC, significantly outperforming MobileNetV3 (84.0%) and the lower-performing MSVP (67.5%).

On FetalDB, it attains 90.6% accuracy, surpassing the runner-up MSVP (86.6%) and other baselines by a clear margin. On CPN, the model reaches near-perfect performance (99.2% accuracy, 0.999 AUC), matching StarNet and exceeding MSVP (93.8%).

Regarding the MedMNIST evaluation, Liquid-Sequencer secures the highest accuracy on BloodMNIST (96.4%) and BreastMNIST (86.5%), notably outperforming MSVP in the latter (63.6%). While MSVP demonstrates superior metrics on PathMNIST (0.985 AUC), Liquid-Sequencer still maintains the highest accuracy (84.4%). Finally, on PneumoniaMNIST, our model achieves the top accuracy (88.3%) and F1 score (87.2%), confirming its robustness across diverse medical imaging scenarios.

4) *Ablation Study: Effectiveness of Bidirectional LNN*. To validate the effectiveness of the Bidirectional Liquid Neural Network (Bidirectional LNN) module proposed in our model, we conduct an ablation study on the BloodMNIST dataset. Specifically, we design three distinct model configurations for comparison: (1) the original model equipped with the full bidirectional LNN; (2) a unidirectional variant where the Vertical

TABLE IV
ABLATION STUDY OF THE BIDIRECTIONAL LIQUID NEURAL NETWORK

Method	Accuracy
Bidirectional	96.4
Horizontal Only	95.6
Vertical Only	95.5

TABLE V
ABLATION STUDY OF THE LNN MODULE

Module	Accuracy	Parameters
CNN	89.85	672K
LSTM	21.32	588K
RNN	19.46	698K
LNN	96.4	309K

component is set to all zeros, retaining only the horizontal LNN; and (3) a unidirectional variant where the Horizontal component is set to all zeros, retaining only the vertical LNN. By comparing the performance of these configurations, we aim to quantitatively assess the contribution of bidirectional information flow to the model's final classification capability.

Table IV presents the results. The complete bidirectional configuration achieves the highest accuracy of 96.4%. in contrast, restricting the model to a single direction causes a performance decline, with horizontal-only and vertical-only variants yielding 95.6% and 95.5%, respectively. The similarity between the unidirectional results suggests that feature extraction from both dimensions is equally critical. These findings confirm that the bidirectional design effectively integrates orthogonal information, creating a synergistic effect essential for optimal classification.

Comparison with Standard Modules. To assess the LNN module's superiority, we replace it with dimension-matched ($\mathbb{R}^{D \rightarrow D}$) CNN, LSTM, and RNN baselines while keeping the remaining architecture unchanged. As shown in Table V, the LNN module achieves the best performance (96.4% accuracy) with high efficiency (309K parameters). While the CNN baseline performs second best (89.85%), it lags significantly in accuracy and requires over double the parameters (672K). Traditional sequence models (LSTM, RNN) perform poorly ($\sim 20\%$ accuracy), failing to capture complex spatial dependencies. This confirms that the LNN's unique dynamical system design offers a superior balance of feature extraction capability and parameter efficiency compared to conventional deep learning modules.

Evaluation of the DPSE Block. We evaluate the Dual-Path Squeeze Excitation (DPSE) Block against a standard CNN feature extractor and a Vision Transformer-style Patch Embedding layer (Table VI). The DPSE Block achieves the optimal trade-off between performance and complexity. Compared to the CNN-based baseline (96.19% accuracy, 973K parameters), the DPSE model attains slightly higher accuracy (96.4%) with a nearly threefold reduction in parameters (309K). Conversely, while the Patch Embedding model is lightweight (264K), its accuracy drops significantly to 93.13%, indicating a loss of fine-grained local information. Thus, the DPSE Block effectively preserves local inductive biases while maintaining high parameter efficiency.

TABLE VI
ABLATION STUDY OF THE DPSE BLOCK

Module	Accuracy	Parameters
CNN-based	96.19	973K
PatchEmbed-based	93.13	264K
DPSE Block	96.4	309K

5) Robustness Analysis: To evaluate model stability under input disturbances, we design 14 perturbation configurations (A–N) applied to the BloodMNIST test set. These include four levels of Gaussian noise (A–D, $\sigma \in [0.05, 0.3]$), four levels of random occlusion (E–H, $\eta \in [0.02, 0.40]$), and six composite scenarios (I–N) combining both noise and occlusion in symmetric and asymmetric intensities. We compare our model against EfficientNet, ShuffleNet, MobileNet, StarNet, and MSVP. The results are summarized in Figure 5.

Our model demonstrates superior robustness, achieving the highest classification accuracy in 13 out of 14 scenarios. Under single Gaussian noise perturbations (A–D), it consistently leads across all intensity levels. Notably, in heavy (C) and severe (D) noise conditions, the model maintains accuracies of 84.61% and 58.9%, respectively, exhibiting a smoother degradation curve than competitors and confirming strong resistance to pixel-level disturbances.

Regarding structural information loss via occlusion (E–H), our model outperforms baselines in three out of four cases. While ShuffleNet marginally leads in medium occlusion (F) with 85.65%, our model dominates under large (G) and very large (H) occlusion, achieving 69.22% and 43.79%, respectively. This highlights its capability to recognize features despite significant local information loss.

The model's advantage is most pronounced in composite perturbation scenarios (I–N), which approximate complex real-world conditions. It ranks first in all six combined settings. In the noise-dominant scenario (M), it achieves 66.27% accuracy, surpassing the second-best StarNet (63.84%). In the occlusion-dominant scenario (N), it reaches 51.80%, significantly outperforming the runner-up MSVP (43.70%) and ShuffleNet (35.92%). These results confirm that our proposed architecture sustains high reliability across diverse and challenging environmental conditions.

6) Model Performance Visualization: To analyze feature representation, we perform t-SNE visualization on the BloodMNIST, CPN X-Ray, and CHD-US datasets (Figure 4). In contrast to MobileNetV3 and EfficientNet, which exhibit blurred boundaries and significant category overlap, Liquid-Sequencer produces compact intra-class clusters and well-separated inter-class distributions. Despite this improved separation, minor overlaps persist at the peripheries of certain clusters. These intersection regions likely correspond to "hard samples" with ambiguous visual features or low signal-to-noise ratios, which represent the primary source of the remaining classification errors. Overall, the distinct clustering confirms the model's ability to learn discriminative representations.

Figure 6 presents the confusion matrix for the CHD-US dataset (1,000 subjects). The model achieves an overall diagnostic accuracy of 92.0%. Crucially for clinical screening, it demonstrates a high sensitivity of 94.0% (identifying 470 out

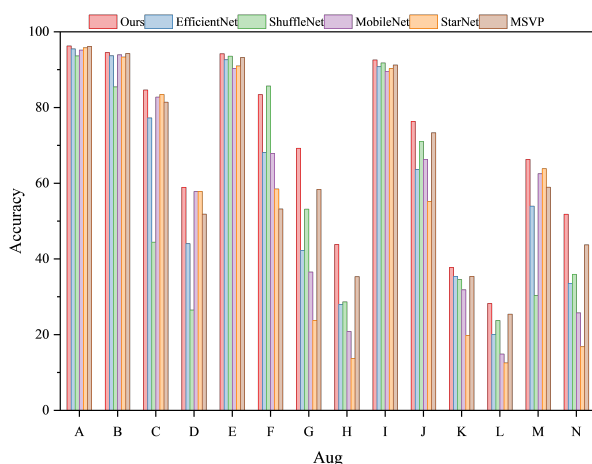


Fig. 5. Experimental Results of Robustness Analysis

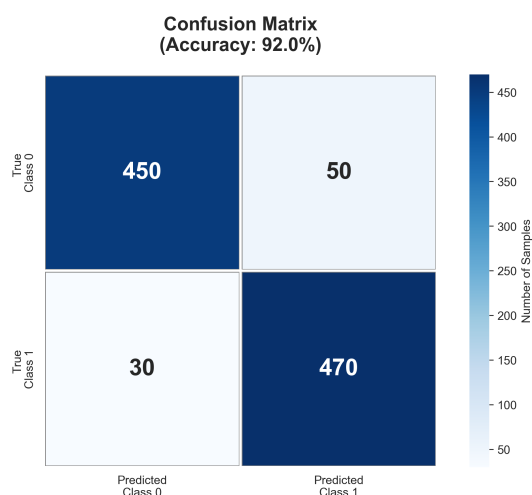


Fig. 6. Confusion matrix of the CHD-US dataset

of 500 positive cases), ensuring a low rate of missed diagnoses (30 False Negatives). Additionally, the model maintains a specificity of 90.0% (correctly classifying 450 out of 500 healthy controls). This performance highlights the model's reliability in distinguishing pathological samples while minimizing false positives.

IV. CONCLUSION

In this study, we propose the Liquid-Sequencer, a lightweight architecture integrating convolutional neural networks with liquid neural networks to capture spatiotemporal features for fetal CHD diagnosis. The model balances high diagnostic accuracy with computational efficiency, achieving expert-level performance suitable for resource-constrained environments. t-SNE visualizations further validate its capability to learn discriminative representations.

Clinically, Liquid-Sequencer offers significant value for prenatal screening, particularly in primary care settings or remote regions lacking specialized hardware. By facilitating early CHD detection, it potentially reduces mortality rates through timely intervention. Additionally, its lightweight design supports efficient inference in digital twin systems. How-

ever, a limitation remains in processing ultra-high-resolution video streams in real-time, where latency may increase on edge devices with extremely limited processing power. Future research will address this by optimizing the liquid neural network's ODE solvers for faster convergence and validating generalization across larger, multi-center datasets.

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